Overblik over Hesel ligningerne

s224041

March 2026

1 Standard form

HESEL ligningerne er givet på følgende form:

$$\mathrm{d}_t n + n\mathcal{K}(\phi) - \mathcal{K}(p_e) = \Lambda_n \tag{1}$$

$$\mathrm{d}_t^0 \omega^* + \{\nabla\phi, \nabla p_i\} - \mathcal{K}(p_e + p_i) = \Lambda_\omega \tag{2}$$

$$\frac{3}{2}\mathrm{d}_t p_e + \frac{5}{2} p_e \mathcal{K}(\phi) - \frac{5}{2}\mathcal{K}\left(\frac{p_e^2}{n}\right) = \Lambda_{p_e} \tag{3}$$

$$\frac{3}{2}\mathrm{d}_t p_i + \frac{5}{2} p_i \mathcal{K}(\phi) + \frac{5}{2}\mathcal{K}\left(\frac{p_i^2}{n}\right) - p_i\mathcal{K}(p_e + p_i) = \Lambda_{p_i} \tag{4}$$

2 Lambda termer

Lambda termerne er givet ved:

$$\Lambda_n = D_e \nabla_\perp^2 n - \frac{n}{\tau} - \alpha\left(\tilde{T}_e + \frac{\tilde{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) - \frac{n - n_p}{\tau_p}$$

$$\Lambda_\omega$$

3 Operatorer

Herved er operatorerne defineret med:

$$\mathrm{d}_t^0 f = \partial_t f + \{\phi, f\}$$

$$\mathrm{d}_t f = \partial_t f + \frac{1}{B}\{\phi, f\}$$

$$\{f, g\} = \partial_x f \partial_y g - \partial_y f \partial_x g$$

$$\mathcal{K}(f) = -\frac{\rho_s}{R}\partial_y f$$

$$\nabla = (\partial_x, \partial_y, \partial_z)^T$$

$$\nabla_\perp = (\partial_x, \partial_y, 0)^T$$

4 Funktioner

Funktionerne er gevet ved:

$$n(x,y,t) : "Tæthed"$$
$$p_e(x,y,t) : "Elektrontryk"$$
$$p_i(x,y,t) : "Iontryk"$$
$$\phi(x,y,t) : "Elektriskpotentiale"$$

5 Udtryk

Samling af udtryk:

$$\phi^*(x,y,t) = \phi(x,y,t) + p_i(x,y,t)$$
$$\rho_s = \sqrt{\frac{T_{e0}}{m_i \Omega_{ei}^2}}$$